The Taste-o-tron

Your AI Powered kitchen assistant

Team Rocket Green Borer Data Solutions

The Taste-o-tron's 4 Ingredients

GAN

Creates a segmentation mask to recognize food vs not food in an image

AE

Cleans the segmentation mask

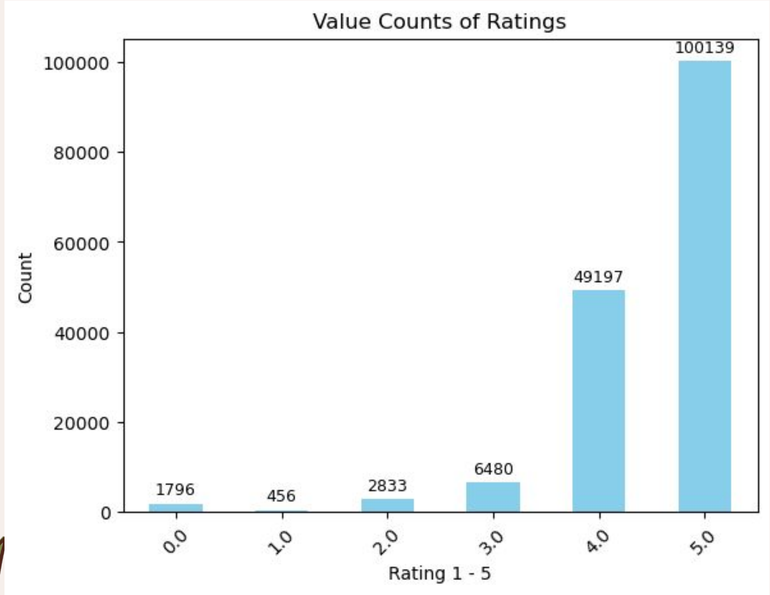
CNN

Classifies the food in an image

NLP

Predicts flavor level (1-5) and describes flavor profile

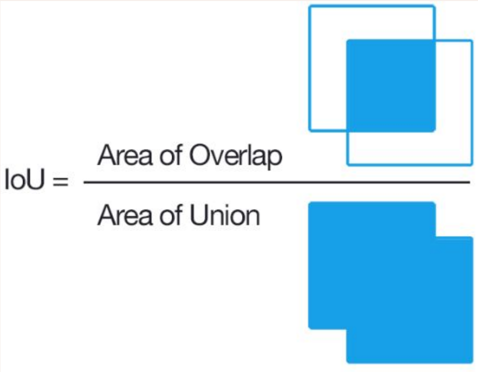
Performance - NLP



- NLP food rating performed pretty low, around 0.62 accuracy
  - Imbalanced data set
- NLP text generation did not produce meaningful text

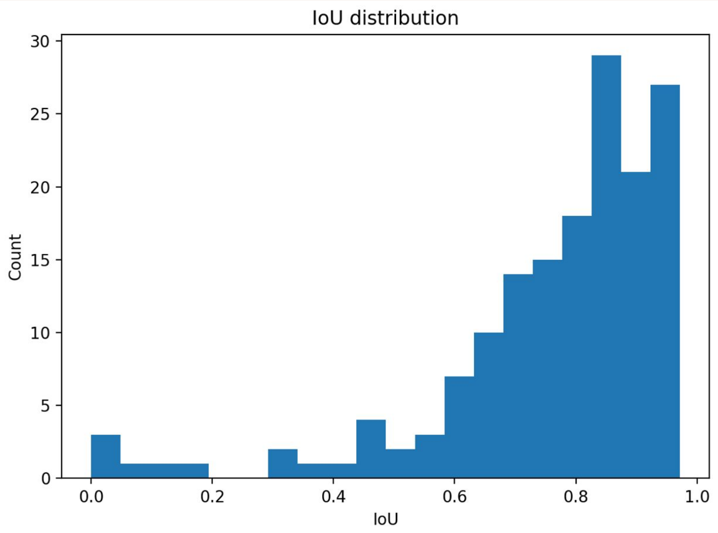
Challenges

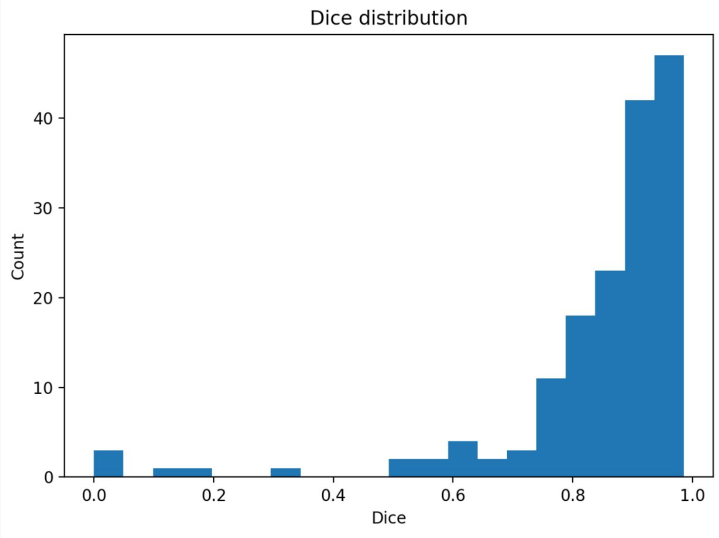
- File structure troubles
  - Managing large dataset
  - VScode and Colab workflow
- Integrating all the models together
  - Time constraints
  - CNN training time really long - only used 10% of data
  - AE used dummy GAN



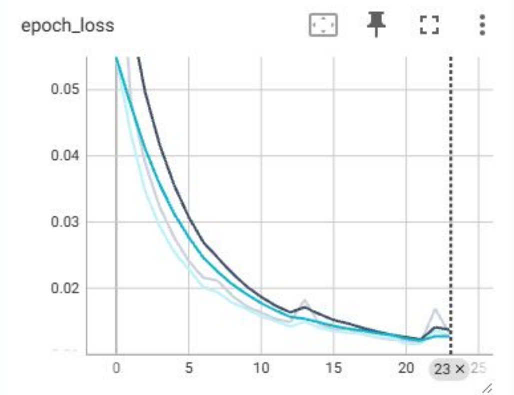
Performance - GAN

Jaccard IoU = Intersection/Union
Jaccard Index values ~0.60 to ~0.95, anything above ~0.75 is very good
Dice takes into account total area of food, shows excellent performance

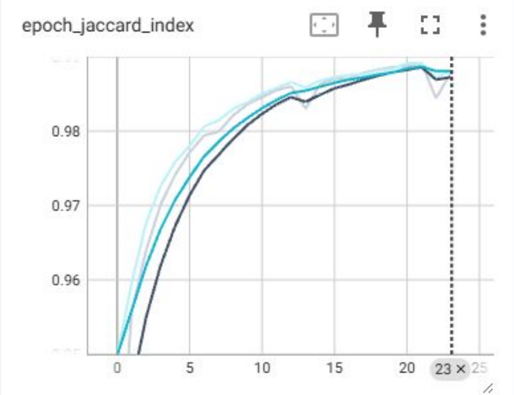




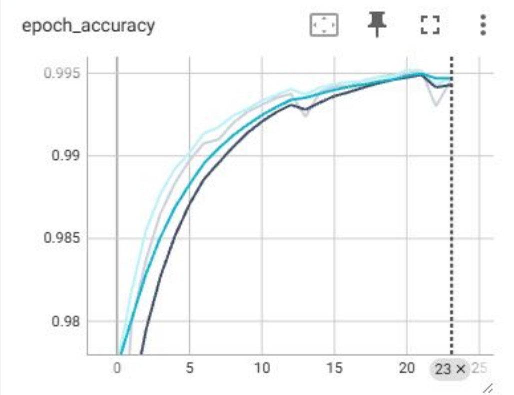
Performance - Autoencoder



Run ↑	Smoothed	Value	Step	Relative
train	0.0137	0.0131	23	24.01 min
validation	0.0127	0.0127	23	24.01 min



Run ↑	Smoothed	Value	Step	Relative
train	0.9872	0.9877	23	24.01 min
validation	0.9881	0.9881	23	24.01 min



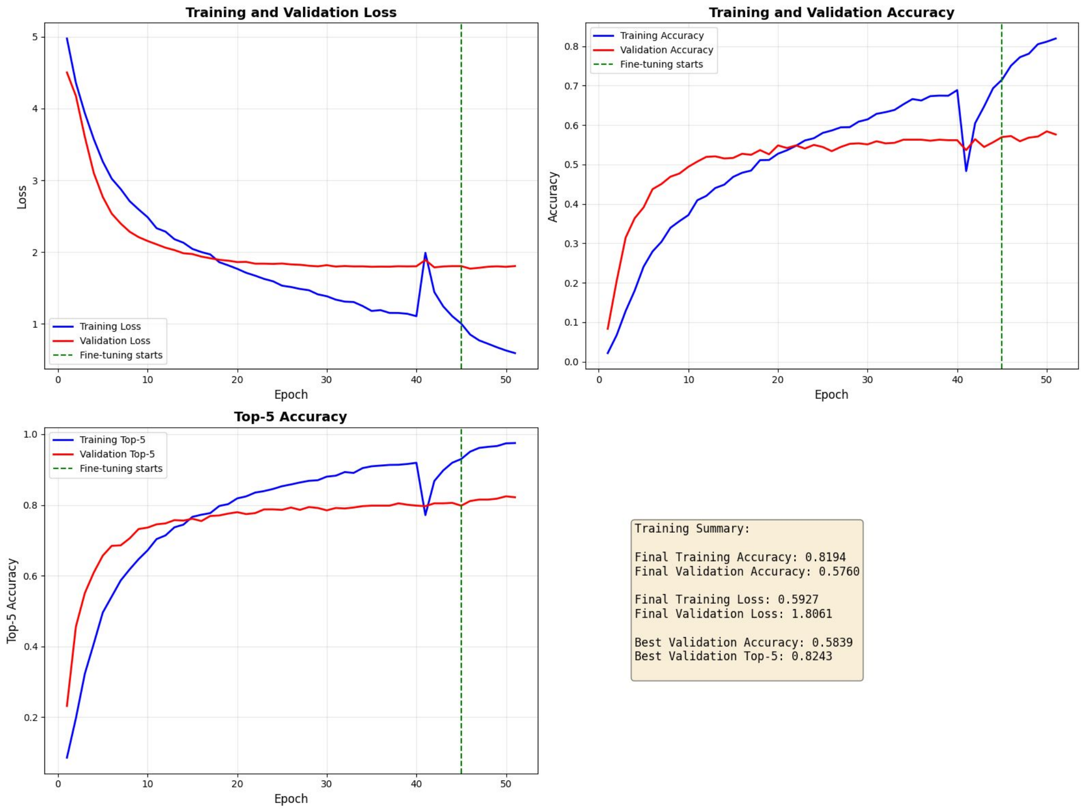
Run ↑	Smoothed	Value	Step	Relative
train	0.9943	0.9945	23	24.01 min
validation	0.9947	0.9947	23	24.01 min

Autoencoder accuracy and Jaccard index is high, what does this mean?

similarity between input and output mask vs similarity between the entire image. how high should this be?

Performance - CNN Training

CNN accuracy only hit 60% without a mask (87.25% for top 5 categories), but with a GAN/Mask it jumped to 97% accuracy for our test example



Demo

Application Notebook -

https://github.com/jacqhorizon/csb410_final/blob/main/notebooks/5_application.ipynb